

SAMPLE COURSE OUTLINE

Teaching Science in the Elementary Grades — Chemistry & Biology

Course Information

Course Title: Teaching Science in the Elementary Grades (Chemistry and Biology)

Course Code: Sci 1

Credit Units: 3 units

Prerequisite: None

Course Description: An introductory professional-education course that builds future teachers' grasp of core Chemistry and Biology concepts as they appear across the elementary grade levels, while developing skill in planning lessons, designing instructional materials, and assessing science learning. The Chemistry strand covers the properties, structure, and changes of matter; the Biology strand covers plant and animal structures and functions, heredity and variation, biodiversity and evolution, and ecosystems.

Intended Learning Outcomes

By the end of the course, students should be able to:

- Explain the K to 12 science curriculum framework and its spiral progression
- Demonstrate mastery of foundational Chemistry and Biology concepts at the elementary level
- Apply a range of teaching strategies suited to elementary science instruction
- Design instructional materials appropriate to specific topics and grade levels
- Use technology to support creative and innovative science teaching
- Select and apply suitable tools for assessing student learning in science
- Model scientific inquiry and sound professional values while teaching science

Weekly Course Outline

Week	Topics / Content	Learning Focus	Activities / Output
Week 1	Classroom Orientation (Setting of expectation, Self-introduction, Classroom Policies, VMGO, Introduction to the Course syllabus) Introduction to the course: Teaching Science in the Elementary Grades: An Overview • Lesson 1: The Science Framework in K to 12 • Lesson 2: Contents of Elementary Science in a Spiral Progress • Lesson 3: The Teaching of Science in the Elementary Grades	Introduce the K to 12 science framework and explain how content spirals across grade levels	Class discussion; short reflection on personal experiences learning science

Week	Topics / Content	Learning Focus	Activities / Output
Week 2	Module 1: Properties of Matter • Lesson 1: Solid, Liquid and Gas (Grade 3) • Lesson 2: Grouping Materials Based on Properties (Grade 4)	Grade 3–4 entry points: solids/liquids/gases; classifying materials by properties	Hands-on sorting activity; worksheet
Week 3	Module 1: Properties of Matter • Lesson 3: Physical and Chemical Properties of Matter: Useful and Harmful Materials (Grade 5) • Lesson 4: Mixtures and Their Characteristics (Grade 6)	Grade 5–6 entry points: distinguishing useful/harmful materials; describing mixtures	Small-group reporting; concept map
Week 4	Module 2: Changes that Materials Undergo (Grade 3) • Lesson 1: Changes that Materials Undergo (Grade 3) • Lesson 2: Changes that Materials Undergo: Useful and Harmful (Grade 4)	Grade 3–4 entry points: identifying changes; useful vs. harmful effects	Group discussion; worksheet
Weeks 5–6	Module 2: Changes that Materials Undergo (Grade 3) • Lesson 3: Changes that Materials Undergo due to Oxygen and Heat (Grade 5) • Lesson 4: Separating Mixture (Grade 6)	Grade 5–6 entry points: causes of change; separation techniques	Demonstration; lab-style worksheet
Week 7 (2nd Trinal)	Module 1: Parts and Functions of Human Being • Lesson 1: Human Sense Organs (Grade 3) • Lesson 2: Human Major Body Organs (Grade 4)	Grade 3–4 entry points: sense organs and their care; major body organs	Group presentation
Week 8	Module 1: Parts and Functions of Human Being • Lesson 3: Male and Female Reproductive Systems (Grade 5)	Grade 5–6 entry points: reproductive system (age-appropriate); organ systems working together	Short research task on medical advances

Week	Topics / Content	Learning Focus	Activities / Output
	• Lesson 4: The Human Organ System at Work (Grade 6)		
Week 9	Module 2: Heredity, Inheritance and Variations • Lesson 1: Living Things Reproduce (Grade 3) • Lesson 2: Life Cycles of Human, Animals and Plants (Grade 4)	Grade 3–4 entry points: how living things reproduce; life cycles	Group discussion; illustration
Week 10	Module 2: Heredity, Inheritance and Variations • Lesson 3: Reproduction Among Flowering Plants (Grade 5) • Lesson 4: Reproduction in Non-Flowering Plants (Grade 6)	Grade 5–6 entry points: flowering plant reproduction; variation among organisms	Group discussion
Week 11	Module 3: Biodiversity and Evolution • Lesson 1: Animals and Plants: Parts, Functions and Importance to Humans (Grade 3) • Lesson 2: Plants and Animals in their Habitats (Grade 4)	Grade 3–4 entry points: plant and animal parts, purposes, and environmental adaptations	Group discussion; classification activity
Week 12	Module 3: Biodiversity and Evolution • Lesson 3: Reproductive Structures of Animals and Plants (Grade 5) • Lesson 4: Common Characteristics of Animals and Plants for Classification (Grade 6)	Grade 5–6 entry points: structural classification and reproductive anatomy of animals and plants	Peer review session; classification matrix
Week 13 <i>(3rd Trinal)</i>	Module 4: Ecosystem • Lesson 1: Living things depend on their environment for basic needs (Grade 3) • Lesson 2: Beneficial and Harmful Interactions among living things (Grade 4)	Grade 3–4 entry points: living things and their environment; beneficial/harmful interactions	Research presentation on a natural cycle
Week 14	Module 4: Ecosystem • Lesson 3: Interactions in estuaries	Grade 5–6 entry points: estuaries, intertidal zones, coral reefs, rainforests	Community-based project proposal

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	and Intertidal Zones (Grade 5) • Lesson 4: Interactions Among living things in Coral Reefs and Tropical Rainforest (Grade 6)		
Week 15	Applying the Content: Lesson planning practice (Chemistry strand, Grades 3–4)	Translating spiral-curriculum content into a classroom lesson plan	Draft lesson plan
Week 16	Applying the Content: Lesson planning practice (Chemistry strand, Grades 5–6)	Continued lesson-plan development and peer feedback	Draft lesson plan
Week 17	Applying the Content: Lesson planning practice (Biology strand, Grades 3–4)	Translating human-body and ecological content into a classroom lesson plan	Draft lesson plan
Week 18	Applying the Content: Lesson planning practice (Biology strand, Grades 5–6); Course wrap-up	Finalizing lesson plans; portfolio compilation	Final lesson plan portfolio